

Task1 Marks-3	<u>Shift Left k Cells</u> Consider an array named source. Write a method/function named shiftLeft(source, k) that shifts all the elements of the source array to the left by 'k' positions. You must execute the method by passing an array and number of cells to be shifted. After calling the method, print the array to show whether the elements have been shifted properly. Example: source=[10,20,30,40,50,60] shiftLeft(source,3) After calling shiftLeft(source,3), printing the array should give the output as: [40, 50, 60, 0, 0, 0]
Task2 Marks-2	<u>Shift Right k Cells</u> Consider an array named source. Write a method/function named shiftRight(source, k) that shifts all the elements of the source array to the right by 'k' positions. You must execute the method by passing an array and number of cells to be shifted. After calling the method, print the array to show whether the elements have been shifted properly. Example: source=[10,20,30,40,50,60] shiftRight(source,3) After calling shiftRight(source,3), printing the array should give the output as: [0,0,0,10,20,30]
Task3 Marks-3	<u>Rotate Left k cells</u> Consider an array named source. Write a method/function named rotateLeft(source, k) that rotates all the elements of the source array to the left by 'k' positions. You must execute the method by passing an array and number of cells to be shifted. After calling the method, print the array to show whether the elements have been shifted properly. Example: source=[10,20,30,40,50,60] rotateLeft(source,3) After calling rotateLeft(source,3), printing the array should give the output as: [40, 50, 60, 10, 20, 30]
Task4 Marks-2	<u>Rotate Right k cells</u> Consider an array named source. Write a method/function named rotateRight(source, k) that rotates all the elements of the source array to the right by 'k' positions. You must execute the method by passing an array and number of cells to be shifted. After calling the method, print the array to show whether the elements have been shifted properly. Example: source=[10,20,30,40,50,60] rotateRight(source,3) After calling rotateRight(source,3), printing the array should give the output as: [40, 50, 60, 10, 20, 30]
Task5	<u>Remove an element from an array</u>

Marks-3	Consider an array named source. Write a method/function named remove(source, size, idx) that removes the element in index idx of the source array. You must execute the method by passing an array, its size and the idx(that is the index of the element to be removed). After calling the method, print the array to show whether the element of that particular index has been removed properly. Example: source=[10,20,30,40,50,0,0] remove(source,5,2) After calling remove(source,5,2) , printing the array should give the output as: [10,20,40,50,0,0,0]
Task 7 Marks-4	<u>Remove all occurrences of a particular element from an array</u> Consider an array named source. Write a method/function named removeAll(source, size, element) that removes all the occurrences of the given element in the source array. You must execute the method by passing an array, its size and the element to be removed. After calling the method, print the array to show whether all the occurrences of the element have been removed properly. Example: source=[10,2,30,2,50,2,2,0,0] removeAll(source,7,2) After calling removeAll(source,7,2), all the occurrences of 2 must be removed. Printing the array afterwards should give the output as: [10,30,50,0,0,0,0,0,0]
Task 8 Marks-4	<u>Repetition</u> Write a method that takes in an array as a parameter and counts the repetition of each element. That is, if an element has appeared in the array more than once, then its 'repetition' is its number of occurrences. The method returns true if there are at least two elements with the same number of 'repetition'. Otherwise, return false. Input: {4,5,6,6,4,3,6,4} Output: True Explanation: Two numbers repeat in this array: 4 and 6. 4 has a repetition of 3, 6 has a repetition of 3. Since two numbers have the same repetition output is True. Input: {3,4,6,3,4,7,4,6,8,6,6} Output: False Explanation: Three numbers repeat in this array: 3, 4 and 6. 3 has a repetition of 2, 4 has a repetition of 3, 6 has a repetition of 4. Since no two numbers have the same repetition output is False.